

Shape-Dependent Complementary Ditopic Terpyridine Pair with Two Levels of Self-Recognition for Coordination-Driven Self-Assembly

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Molecular recognition in biological systems plays a vital role in the precise construction of biomacromolecules and the corresponding biological activities. Such recognition mainly relies on the highly specific binding of complementary molecular pairs with complementary sizes, shapes, and intermolecular forces. It still remains challenging to develop artificial complementary motif pairs for coordination-driven self-assembly. Herein, a series of shape-dependent complementary motif pairs, based on ditopic 2,2':6',2''-terpyridine (TPY) backbone, are designed and synthesized. The fidelity degrees of self-assemblies from these motifs are carefully evaluated by multi-dimensional mass spectrometry, nuclear magnetic resonance spectroscopy, and molecular modeling. In addition, two levels of self-recognition in both homoleptic and heteroleptic assembly are discovered in the assembled system. Through finely tuning the shape and size of the ligands, a complementary pair is developed with error-free narcissistically self-sorting at two levels of self-recognition, and the intrinsic principle is carefully investigated.

based on the complementarity of size, shape, and intermolecular forces, is widespread in biological systems.^[1] It plays vital roles in most of the biological activities, involving the complementary pairs of nucleobases applied in DNA for storage and transmission of genetic information,^[2,3] antigen-antibody for the regulation of immune response,^[4] sugar-lectin for intercellular recognition,^[5] enzyme-substrate for catalysis,^[6] etc. Beyond that, the complementary pair of streptavidin-biotin known in biological systems,^[7] has been extensively applied in biosensing and separation owing to its strong binding affinity, high specificity, and stability.^[8]

In artificial systems, the development of synthetic complementary motif pairs inspired by the biological ones has received considerable attention over the past few decades,^[9] due to the prospect that the

defect-free self-sorting among a dynamic combinatorial library^[10,11] would advance the construction of structures and exploration of functions, particularly for supramolecular chemistry. Among the diverse non-covalent interactions, metal-ligand coordination is one of the ideal driving forces for achieving spontaneous and high-fidelity recognition, owing to the highly directional and predictable feature.^[12] Based on coordination-driven self-assembly, a vast variety of discrete^[12] and polymeric structures^[13] have been constructed with high precision, including helicates,^[14] grids,^[15,16] polygons,^[17–19] links,^[20] knots,^[21] fractals,^[22] polyhedrons,^[23–25] etc. However, in a multi-component system without selective molecular recognition, the coordination-driven self-assembly often leads to a statistical mixture of homoleptic and heteroleptic complexes. To prevent the generation of undesired complexes, many efforts have been carried out, including the following strategies: i) coordination nature control via charge separation effect in Pt(II) complexes for reducing the electrostatic repulsion,^[26] or via trigonal bipyramidal coordination geometry of Cu(II) for heteroleptic self-assembly;^[27] ii) guest molecule induction via template effect or host-guest chemistry;^[28] iii) tailoring geometric complementarity based on maximum coordination sites for designed structures, and/or geometric constraints for mismatched complexes with distortion penalties;^[29–33] iv) ligand

1. Introduction

Molecular recognition that refers to the specific binding of two or more molecules in either narcissistic or social self-sorting way

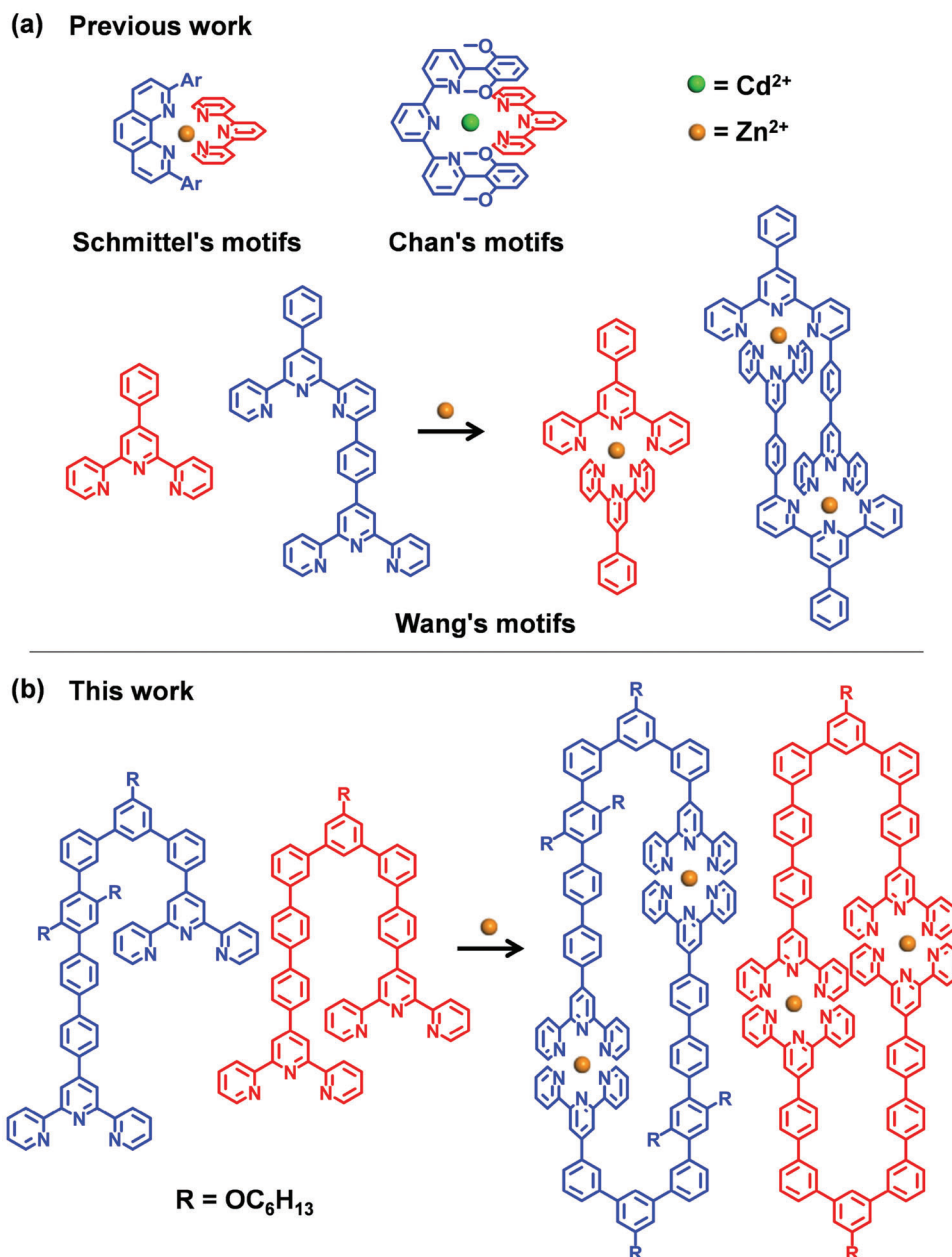
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Scheme 1. a) Substituted terpyridine (TPY)-based complementary motif pairs in previous works. b) Unsubstituted TPY-based shape-dependent complementary motif pair in this work.

modification for enlarging steric hindrance, or/and enhancing ligand-ligand π - π interactions;^[34–38] etc.

In the field of coordination-driven self-assembly based on pyridine, bipyridine, phenanthroline, and terpyridine, however, only a few types of motif pairs have been developed, for instance, introducing methyl groups at 2,6-positions of pyridine,^[34,35] anchoring aromatic groups at 2,9-positions of phenanthroline,^[36] or at 6,6''-position of 2,2':6',2''-terpyridine (TPY),^[37] as well as linking multiple TPY groups in a head to tail way^[38] (as summarized in **Scheme 1a**). One of the major obstacles is the lack of knowledge for precise manipulation of ligand-ligand interactions. Thus, tremendous efforts are often required in screening

a proper moiety or/and suitable positions of ligand for substitution.

Beyond those boundaries, we describe herein a shape-dependent approach for developing complementary building block pairs based on TPY without any substitution (**Scheme 1b**). Four sets of ditopic ligands have been designed and synthesized via systemically alternating the shape and size parameters of the backbones, which finally afforded one pair with error-free narcissistic self-sorting capability. The rigidity of ligand backbones and the shape variances between a ligand pair were recognized as the key factors for determining the fidelity of the self-sorting, in the form of three parameters, that is, width difference between the

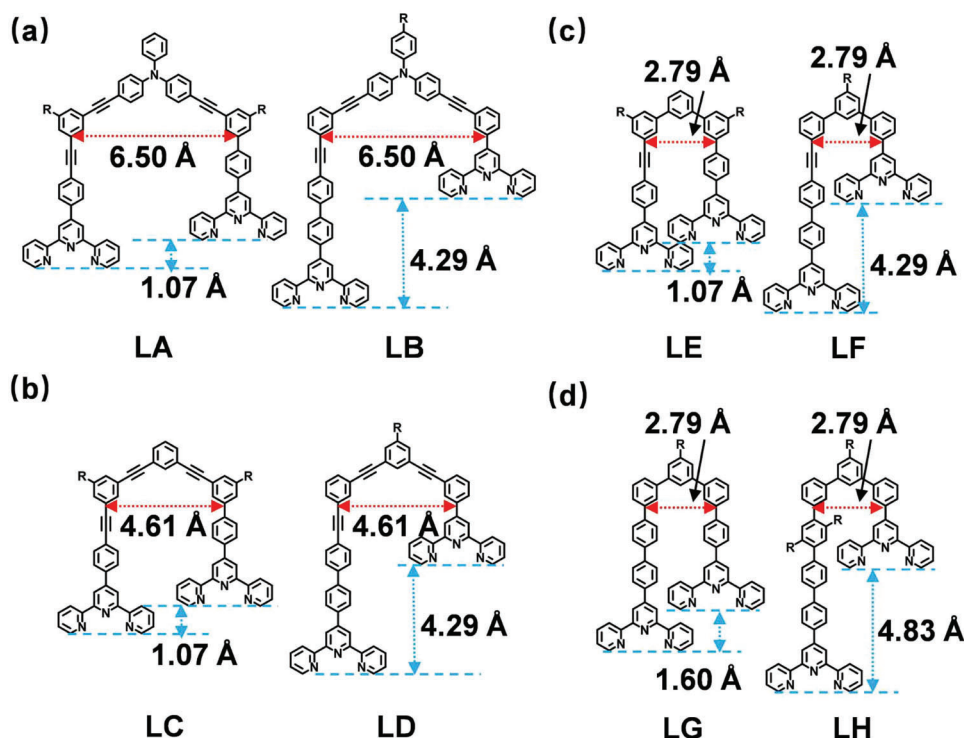


Figure 1. Molecular structures of a) LA and LB, b) LC and LD, c) LE and LF, d) LG and LH with width and length differences between the two arms of ligands measured based on their corresponding geometrically optimized structures, respectively.

two arms, length difference between two arms, and shape difference between a pair of complementary ligands. In addition to gaining insights for developing new complementary motif pairs, the reported shape-dependent complementary motif pair could be further served as building blocks for constructing more sophisticated metallo-supramolecular architectures.

2. Results and Discussion

Aiming to develop shape-dependent complementary motif pairs with self-sorting ability, our design was guided by the following three criteria: i) using rigid motif backbones to enhance torsion energy penalties during the formation of mismatched complexes; ii) enlarging shape discrimination to achieve specific shape-dependent recognition; and iii) constructing identical supramolecular framework using each set of motif pair for further applications in fabricating higher sophisticated architectures with a programmable approach.

Based on these requirements, we first designed the pair of ditopic TPY ligands, LA/LB, as shown in **Figure 1a**. These motifs have fully π -conjugated backbones with triphenylamine moieties (TPA) as the corners, and alkynyl/phenyl groups as the spacers. The length variances of the two arms of LA/LB ($\Delta I_A/\Delta I_B$) were measured as 1.07/4.29 Å, respectively, based on the modeling results (Figure S1, Supporting Information). It is worth noting that the shape difference between LA and LB is 3.22 Å ($\Delta\Delta I = \Delta I_B - \Delta I_A$), almost reaching the maximum value, owing to the possible synthetic difficulties and poor solubility caused by further extending the arm length. Based on the corresponding simulation results (**Figure 2a**), the resultant homolep-

tic metallo-supramolecular cycles, Zn_2LA_2 and Zn_2LB_2 , have the same shapes and chemical compositions of frameworks (regardless of the substituted alkyl chains), but different metal ion locations. Therefore, different numbers of hexyl chains were anchored on the two motifs (2 on LA, and 1 on LB), to distinguish the possible assembled structures, that is, Zn_2LA_2 , Zn_2LALB , and Zn_2LB_2 (Figure 2a) via electron spray ionization-mass spectrometry (ESI-MS).

The syntheses of LA and LB were accomplished by multi-step of Sonogashira and Suzuki cross-coupling reactions with decent yields, and the synthetic details of LA and LB were summarized in Supporting Information (Scheme S1, Supporting Information). The self-assembly was carried out through mixing pure LA, pure LB, or an LA/LB mixture at a molar ratio of 1:1 with $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, respectively, under a stoichiometric ratio in $\text{CHCl}_3/\text{MeOH}$ (v/v = 1:1) at 50 °C for 8 h. After the anions were exchanged to PF_6^- , the so-formed assemblies were directly characterized by ESI-MS for the rapid evaluation of the motif pair complementarity.^[39] Theoretically, without molecular recognition, a statistical distribution of Zn_2LA_2 , Zn_2LALB , and Zn_2LB_2 (molar ratio as 1:2:1) was supposed to be afforded during the assembly of the mixture of LA/LB with $\text{Zn}(\text{II})$; while completely self-sorted homoleptic assemblies would be resulted in the absence of Zn_2LALB . By screening the ESI-MS spectra (Figure 2c), a set of much weaker peaks assigned to the $[\text{Zn}_2\text{LALB}]^{4+}$ species was observed compared to the dominant signals of $[\text{Zn}_2\text{LA}_2]^{4+}$ and $[\text{Zn}_2\text{LB}_2]^{4+}$. Traveling wave ion mobility-mass spectrometry (TWIM-MS)^[40] displayed three signal bands (Figure 2e), corresponding to the three complex species, further indicating the insufficient self-sorting. These results indicated that

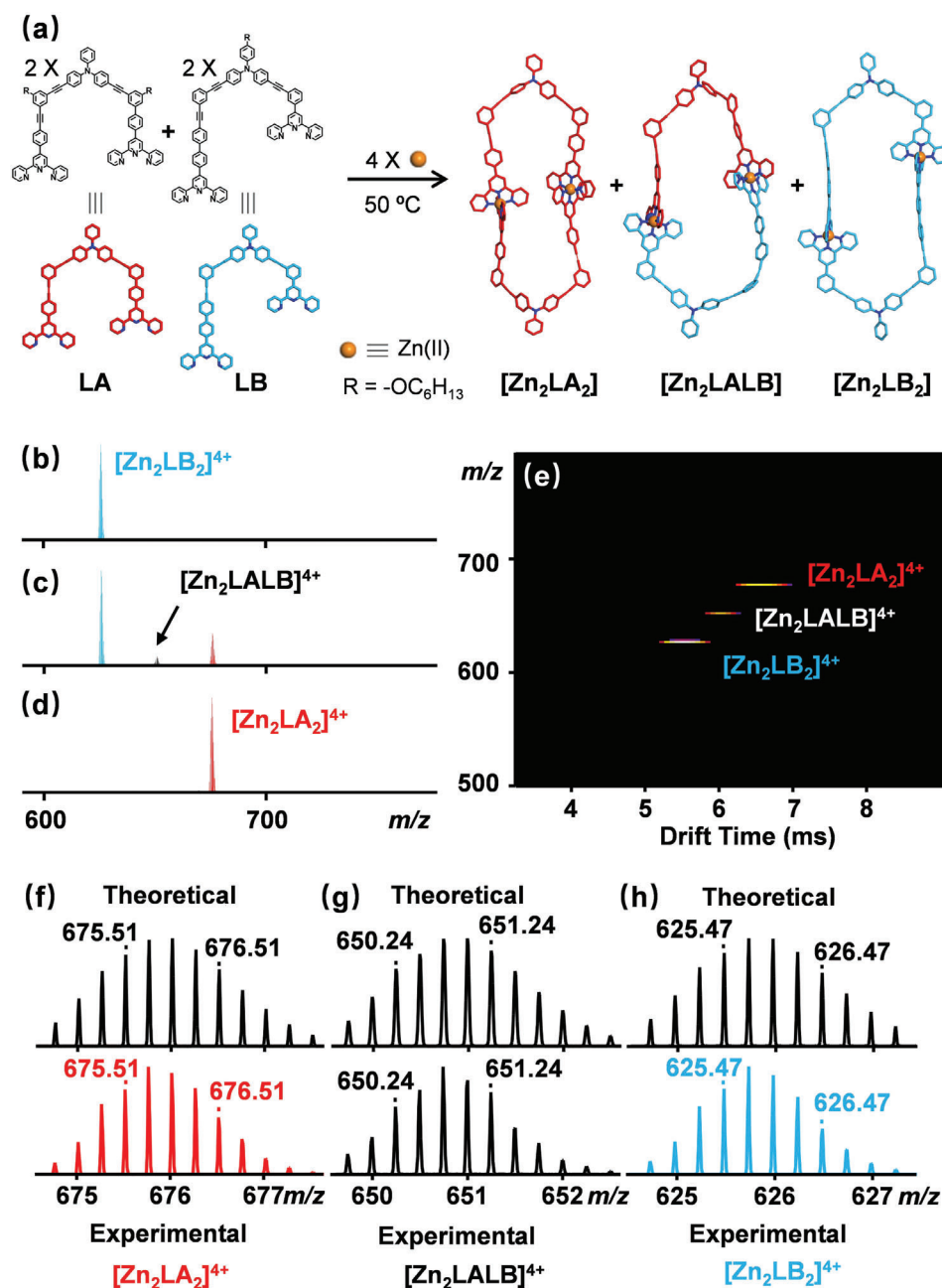


Figure 2. a) Self-assembly of an LA/LB mixture at a molar ratio of 1:1 with Zn(II) (Geometrically optimized structures of the resultant complexes are shown with all alkyl chains being omitted for clarity). ESI-MS spectra of the assembly of b) LB only, c) LA/LB, and d) LA only with Zn(II), respectively, with PF_6^- as the counterion. e) TWIM-MS plots (m/z vs drift time) of the self-assembly of LA/LB with Zn(II). Measured (bottom) and calculated (top) isotope patterns for f) $[Zn_2LA_2]^{4+}$, g) $[Zn_2LALB]^{4+}$, and h) $[Zn_2LB_2]^{4+}$, respectively.

shape-dependent self-recognition partially happened during the complexation based on this pair of motifs with $\Delta\Delta I = 3.41$ Å. It thus encouraged us to modify the motif pair for further enhancing the complementarity. Based on the theoretical simulations of Zn_2LALB (Figure 2a), the distortion caused by the coordination of two shape-mismatched ligands could be released via slightly bending of the TPA moieties.

As such, we reasoned that, increasing the backbone rigidity through shortening the width between the two arms of a pair of

ligands would enlarge the distortion energy penalties from the shape-mismatched assembly with two different ligands. Based on the speculation, we next designed LC and LD (Figure 1b, Figure S2, Supporting Information), via replacing TPA moieties with phenyl groups (see Supporting Information for the detailed synthesis). As a result, the width between two arms was reduced from 6.50 (LA/LB) to 4.61 Å (LC/LD), while $\Delta\Delta I$ of the new pair remained unchanged. However, ESI-MS and TWIM-MS results of the corresponding self-assemblies (Figure 3b–e) still revealed

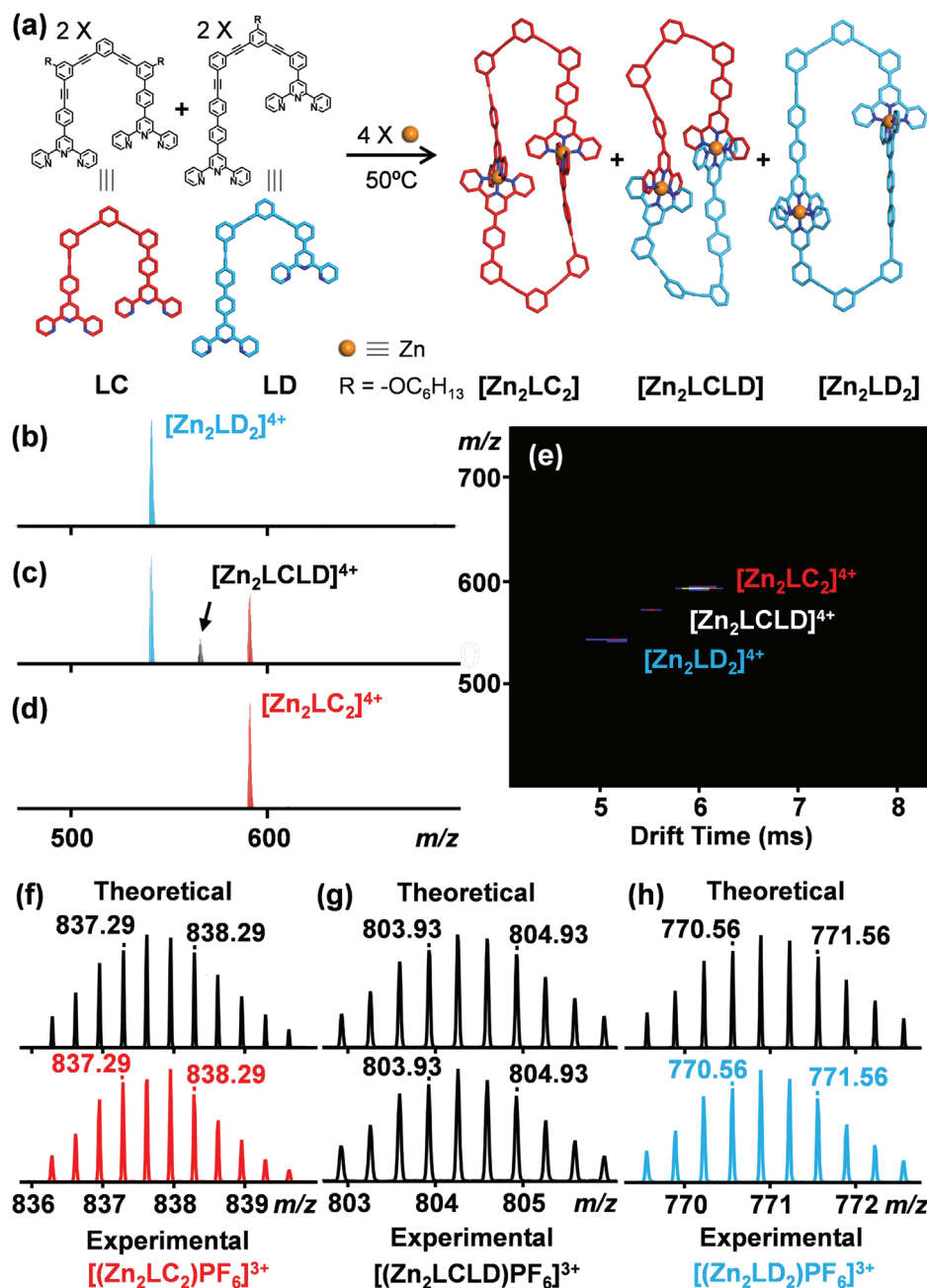


Figure 3. a) Self-assembly of an LC/LD mixture at a molar ratio 1:1 with Zn(II) (Geometrically optimized structures of the resultant complexes are shown with all alkyl chains being omitted for clarity). ESI-MS spectra of the assembly of b) LD only, c) LC/LD, and d) LC only with Zn(II), respectively, with PF₆[−] as the counterion. e) TWIM-MS plots (*m/z* vs drift time) of the self-assembly of LC/LD with Zn(II). Measured (bottom) and calculated (top) isotope patterns for f) [(Zn₂LC₂)PF₆]³⁺, g) [(Zn₂LCLD)PF₆]³⁺, and h) [(Zn₂LD₂)PF₆]³⁺, respectively.

the partial narcissistic self-recognition. Detailed theoretical simulation results suggest that the distortion mainly occurs in the alkynyl regions (Figure 3a).

Further modification was then carried out by removing the “flexible” alkynyl groups attached to the corner phenyl group, and the corresponding ligands, LE/LF, were prepared with the width between their two arms as 2.79 Å (Figure 1c, Figure S3, Supporting Information; the synthesis details were summarized in Supporting Information). As per our expectation, only two sets

of peaks ascribing to Zn₂LE₂ and Zn₂LF₂, respectively, were displayed in the ESI-MS and TWIM-MS spectra for the assembly of a mixture of LE/LF with Zn(II) (Figure 4b–f). Further expanding the ESI-MS in the *m/z* region of [(Zn₂LELF)PF₆]³⁺ exhibited an undetectable signal (Figure 4e), suggesting the high fidelity of the narcissistic self-sorting.

More detailed characterizations about the self-assemblies were carried out via ¹H NMR, and the signals of all ligands and intermediates were assigned based on 2D correlation spectroscopy

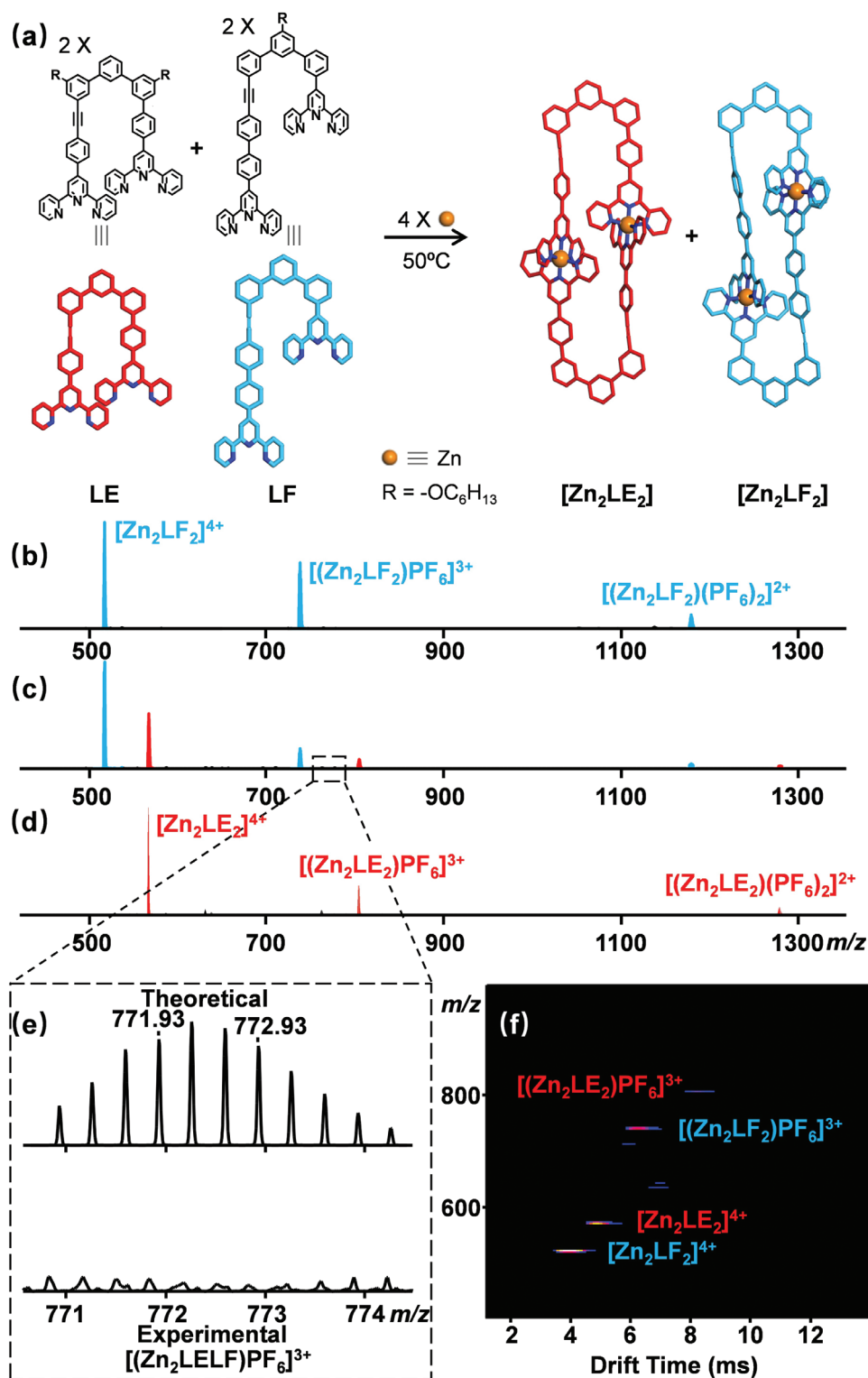


Figure 4. a) Self-assembly of an LE/LF mixture at a molar ratio 1:1 with Zn(II) (Geometrically optimized structures of the resultant complexes are shown with all alkyl chains being omitted for clarity). ESI-MS spectra of the assembly of b) LF only, c) LE/LF, and d) LE only with Zn(II), respectively, with PF_6^- as the counterion. e) Measured (bottom) and calculated (top) isotope patterns for $[(Zn_2LELF)PF_6]^{3+}$. f) TWIM-MS plots (m/z vs drift time) of the self-assembly of LE/LF with Zn(II).

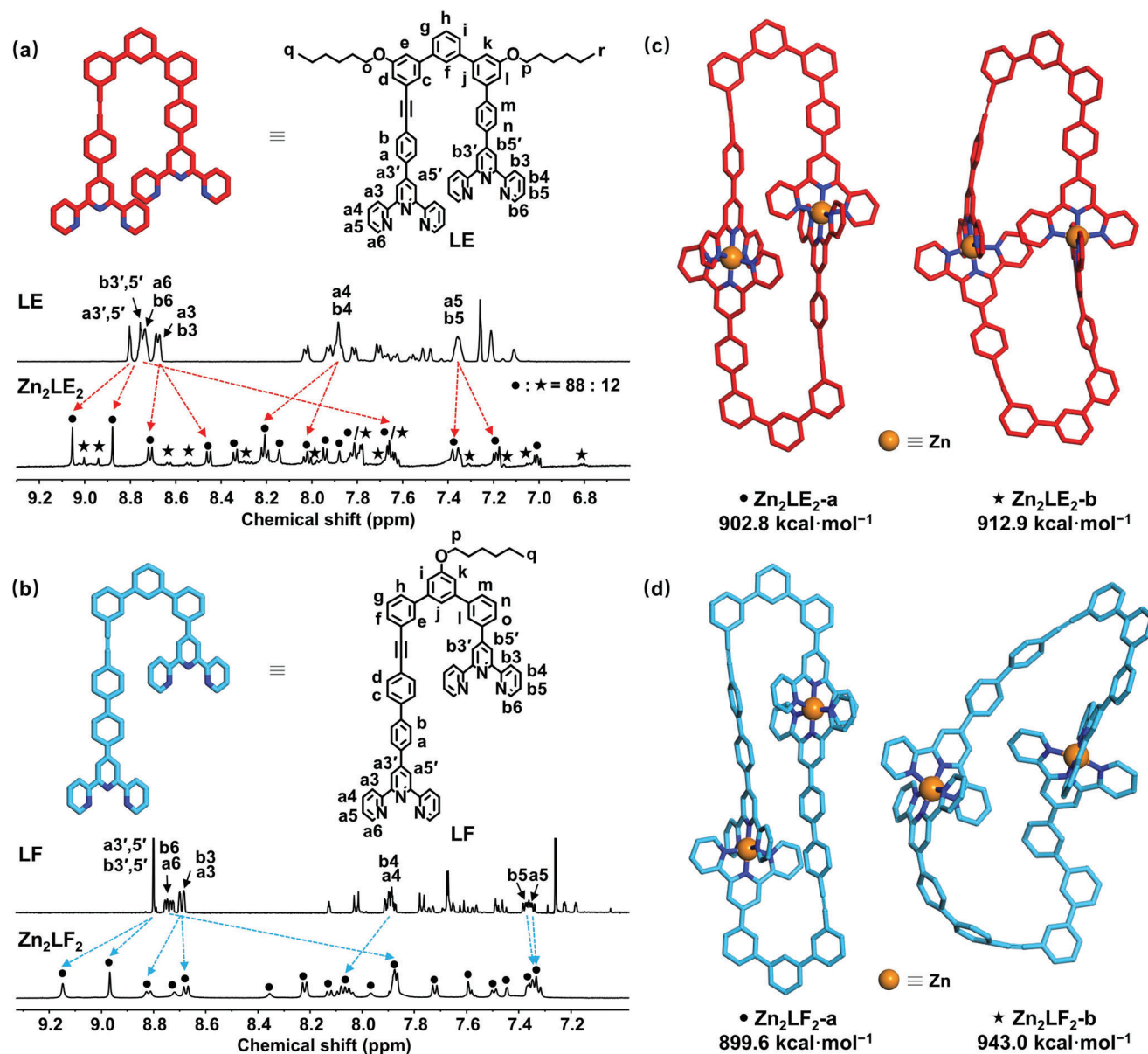


Figure 5. ^1H NMR spectra (500 MHz, 298 K) of a) LE in CDCl_3 and Zn_2LE_2 in CD_3CN ; b) LF in CDCl_3 and Zn_2LF_2 in CD_3CN . Geometrically optimized structures and the corresponding total potential energies of the possible isomers of c) Zn_2LE_2 and d) Zn_2LF_2 . All alkyl chains are omitted for clarity in the model structures.

(2D COSY), and 2D nuclear Overhauser enhancement spectroscopy (2D NOESY) NMR, or rotating-frame Overhauser effect spectroscopy (2D ROESY) NMR (see Supporting Information for all the spectra). Out of our expectation, two distinct sets of protonic resonance signals were displayed in the spectrum of Zn_2LE_2 with an integrated area ratio of 88:12 (Figure 5a), both of which contained all characteristic peaks assigned to the metal-coordinated TPY moieties. It indicated that Zn_2LE_2 was a mixture of two isomers with similar size that could not be resolved even in TWIM-MS. Two isomers of Zn_2LE_2 , denoted as $\text{Zn}_2\text{LE}_2\text{-a/b}$, were proposed based on the theoretical simulation, as shown in Figure 5c. Among them, $\text{Zn}_2\text{LE}_2\text{-a}$ with long-short arm combinations is the desired shape-recognized structure without show-

ing any distortion; while $\text{Zn}_2\text{LE}_2\text{-b}$ with a long-long and short-short arms combination displays some bent alkynyl groups in the simulated structure. We speculated that the existence of isomers came from the small ΔI_E value (1.07 Å). Detailed analysis revealed that the total potential energies of $\text{Zn}_2\text{LE}_2\text{-a/b}$ were estimated as 902.8 and 912.9 kcal·mol $^{-1}$, respectively. The energy gap is much lower than that between $\text{Zn}_2\text{LF}_2\text{-a}$ and $\text{Zn}_2\text{LF}_2\text{-b}$ (10.1 kcal·mol $^{-1}$ vs 43.4 kcal·mol $^{-1}$, Figure 5d). Therefore, no isomer was detected in the ^1H NMR spectrum of Zn_2LF_2 , indicating larger ΔI would prevent the arm-mismatched complex ($\Delta I_F = 4.29$ Å).

Although this set of results based on LE/LF was not as perfectly performed as designed, the mismatching assembly occurred in

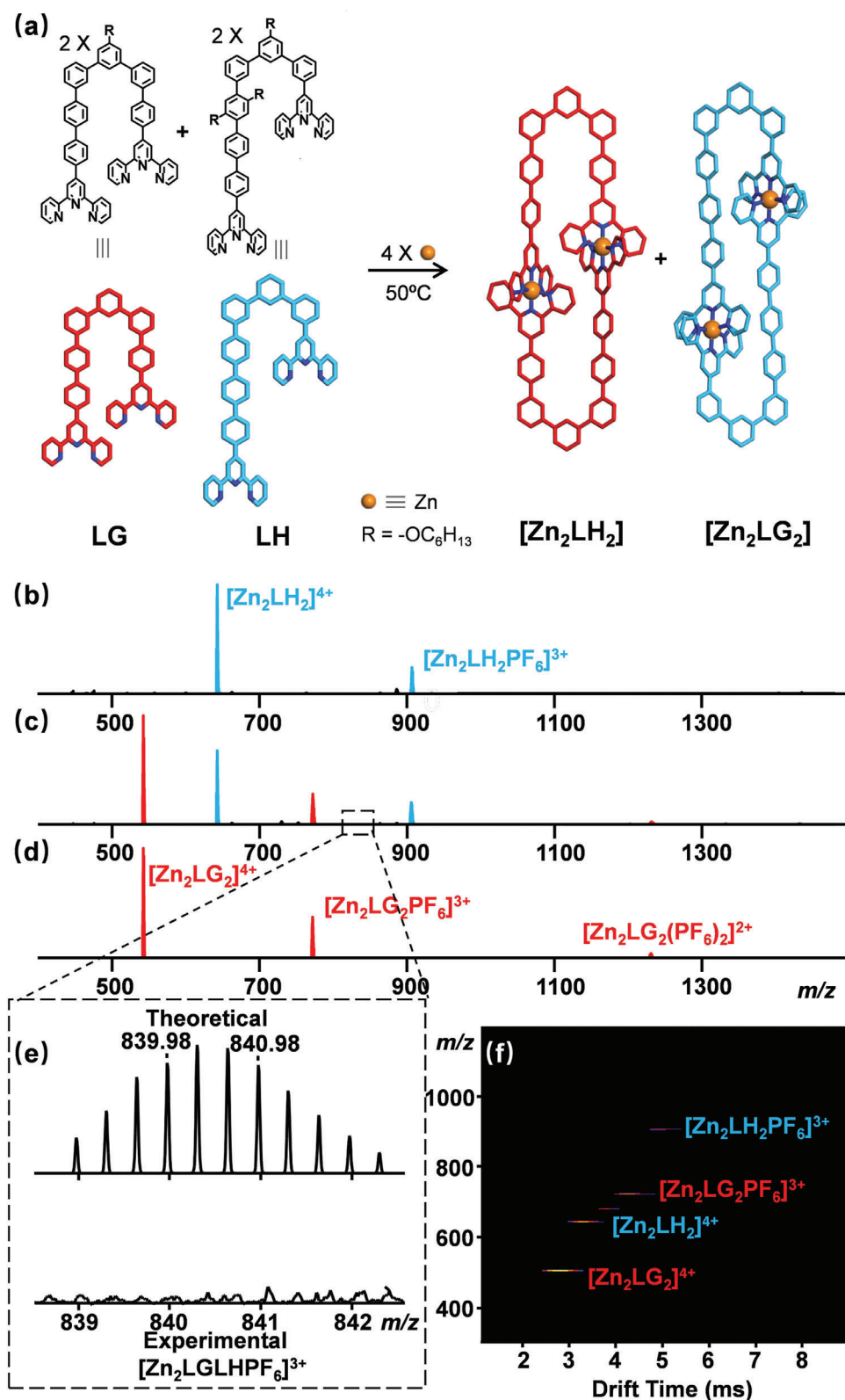


Figure 6. a) Self-assembly of an LG/LH mixture at a molar ratio 1:1 with Zn(II) (Geometrically optimized structures of the resultant complexes are shown with all alkyl chains being omitted for clarity). ESI-MS spectra of the assembly of b) LH only, c) LG/LH, and d) LG only with Zn(II), respectively, with PF_6^- as the counterion. e) Measured (bottom) and calculated (top) isotope patterns for $[Zn_2LGLH]PF_6]^{3+}$. f) TWIM-MS plots (m/z vs drift time) of the self-assembly of LG/LH with Zn(II).

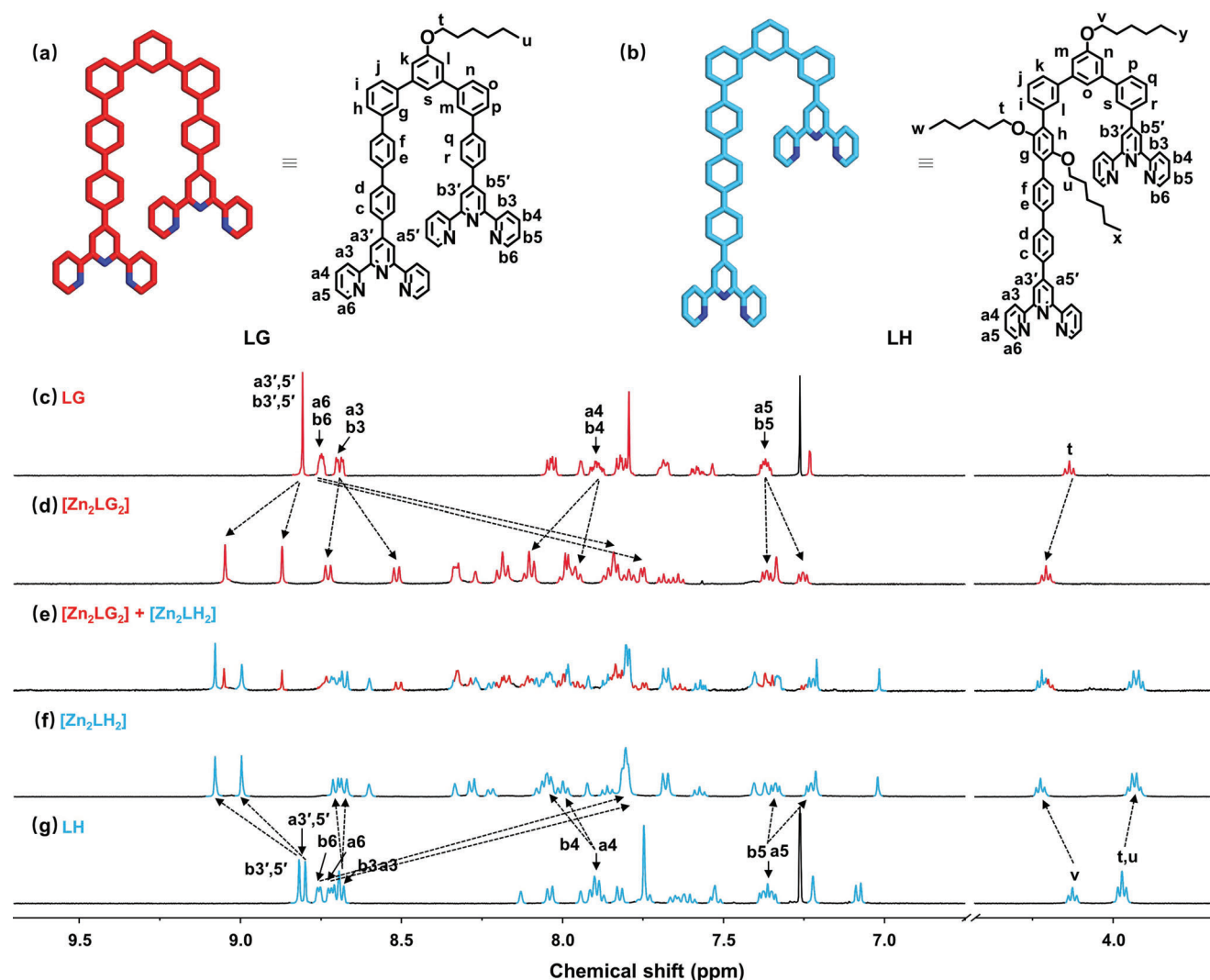


Figure 7. Structures of ligands a) LG and b) LH. ^1H NMR spectra (500 MHz, 298 K) of c) LG in CDCl_3 ; d) Zn_2LG_2 in CD_3CN ; e) an LG/LH mixture (molar ratio 1/1) with $\text{Zn}(\text{II})$ in CD_3CN ; f) Zn_2LH_2 in CD_3CN ; g) LH in CDCl_3 .

the Zn_2LE_2 system reminded us that two levels of molecular recognition need to be considered in our design, that is, isomerization during the homoleptic coordination of two identical ligands and heteroleptic coordination of two different ligands. As a result, we further proposed that extending the ΔI values of the motif pair would achieve shape-dependent self-sorting between different ligands and block isomerization in homoleptic coordination of two identical ligands. Before we moved to the investigation of the next pair of ligands, we first returned back to collect the ^1H NMR spectra of $\text{Zn}_2\text{LA}_2/\text{Zn}_2\text{LC}_2$ systems with the smaller ΔI values also as 1.07 Å. Consistent with the results in Zn_2LE_2 , both the spectra of $\text{Zn}_2\text{LA}_2/\text{Zn}_2\text{LC}_2$ clearly showed two sets of peaks with integrated area ratios of 62:38 (Zn_2LA_2) and 65:35 (Zn_2LC_2), respectively, as shown in Figures S7 and S8, Supporting Information.

To pursue the fully complementary ligand pair, in the design of the fourth pair of ligands, LG/LH (Figure 1d), we kept the corner of ligands as phenyl groups, while replacing the final alkynyl

groups on the longer arms to phenyl rings. Therefore, the width between two arms remains the narrowest at 2.79 Å, $\Delta\Delta I$ is also kept at 3.23 Å, but the ΔI of LG extends to 1.60 Å (Figure 1d, Figure S4, Supporting Information). It was worth noting that, owing to the poor solubility of the longer arm of LH, additional alkyl chains were introduced to increase the solubility.

As shown in the multi-dimensional MS spectra (Figure 6b–f), only the peaks assigned to the homoleptic assemblies, that is, Zn_2LG_2 and Zn_2LH_2 , were observed in the self-assembly of LG/LH mixture with $\text{Zn}(\text{II})$, suggesting the complete self-sorting between LG and LH. To our delight, ^1H NMR spectra of both Zn_2LG_2 and Zn_2LH_2 assembled from their corresponding ligand, respectively, displayed only one set of signals (Figure 7d,f), ruling out the existence of mismatched isomers and suggesting another level of self-recognition in the homoleptic assemblies.

^1H NMR spectrum of the three-component self-assembly (LG/LH with Zn^{2+}) provided additional evidence for the error-free recognition at both homoleptic and heteroleptic assembly

Table 1. Fidelity of the self-assembly of shape-dependent complementary pairs with Zn(II), and the shape parameters of the corresponding motifs.

Motif Pair	Width [Å] ^{a)}	ΔI ^{b)}	$\Delta\Delta I$ ^{c)}	Fidelity [F] ^{d)}
LA/LB	6.50	1.07/4.29	3.22	0.69
LC/LD	4.61	1.07/4.29	3.22	0.75
LE/LF	2.79	1.07/4.29	3.22	0.89
LG/LH	2.79	1.60/4.83	3.23	1.00

^{a)} Measured on the geometrically optimized structures. ^{b)} ΔI is also measured based on the geometrically optimized structures; the two values refer to the two motifs of the listed pair, respectively. ^{c)} $\Delta\Delta I$ is calculated based on the two ΔI values of the pair of motifs. ^{d)} F is determined via integration peak areas of 3",5"-protons of TPY groups, according to the formula: $F = (\text{total amount of desired complexes})/(\text{total amount of all complexes})$.

levels (Figure 7e). Only the signals ascribed to Zn_2LG_2 and Zn_2LH_2 were shown in the spectra without any impurities or isomers, by comparing the spectra of Zn_2LG_2 and Zn_2LH_2 . In addition, benefiting from the well-resolved and characteristic signals of 3",5"-protons of TPY moieties in the spectra, quantitative or semi-quantitative evaluation of the degree-of-fidelity is possible by integrating peak areas of the 3",5"-protons of all species. A parameter of fidelity (F)^[41] was used here for the quantitative evaluation. F value is defined as the mole fraction of the desired complexes (e.g., Zn_2LG_2 -a and Zn_2LH_2 -a) among all complexes (e.g., Zn_2LG_2 -a/b, Zn_2LH_2 -a/b, and Zn_2LGLH , etc., as shown in Figure S10, Supporting Information). Since no undesired complex was found in the self-assembly system of LG/LH with Zn(II), the corresponding F value is 1.00. Based on the method, F values of LA/LB, LC/LD, LE/LF systems were determined as 0.69, 0.75, and 0.89, according to the corresponding ^1H NMR spectra of the three-component self-assembly (Figures S7–S9, Supporting Information), respectively. All the results combined with the size of the ligands were summarized in Table 1. As expected, the width between the arms of ligands, as well as the ΔI and $\Delta\Delta I$ values determine the complementarity of the self-assembly.

To evaluate the dependence of metal ions onto self-recognition, we have performed the self-assembly of a 1:1 mixture of LG/LH with Cd(II), Cu(II) or Fe(II) salts under stoichiometric ratio, respectively. No heteroleptic assemblies were observed in the ESI-MS spectra of all experiments (as shown in Figures S13–S15, Supporting Information), indicating the complete self-recognitions in all cases. The fidelity values of Cd(II) and Fe(II) self-assembled results have been further measured by ^1H NMR, which are also determined as 1.00 (as shown in Figures S11 and S12, Supporting Information). It is worth noting that the ^1H NMR spectrum of Cu(II) assemblies was not satisfactory owing to the paramagnetic feature of Cu(II) complexes. All these results confirm that a wide range of metal ions are applicable for this shape-dependent self-recognition approach.

3. Conclusion

In summary, we have designed and synthesized four pairs of ditopic TPY-based shape-complementary building blocks, by carefully tuning their shape parameters, including the width and the length difference between two arms of a ligand, as well as the shape difference between a pair of ligands. Two levels of recogni-

tions in both homoleptic and heteroleptic assembly were found in the system, owing to the ditopic design of the motif scaffold. After careful investigations, LG/LH with the narrowest width, the largest ΔI and $\Delta\Delta I$ values, displayed the desired error-free self-sorting behavior, which suggests that the backbone rigidity and the shape difference between ligands dominate the molecular recognition. Based on multi-dimensional MS and NMR characterizations, the fidelity degrees of the narcissistically self-sorted complexations have been (semi)-quantitatively evaluated.

In our ongoing works, we will focus on investigating the self-sorting behaviors of more complex systems based on three or more types of ligands with different shapes, sizes, numbers of arms, types of coordination groups, etc., to shed more light on the self-recognition nature in coordination-driven self-assembly, as well as to generate a more general design principle of shape-dependent complementary building blocks. Additionally, more efforts will be focused on utilizing these motifs for the construction of assemblies with higher sophisticated structures and advanced functions.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

coordination-driven self-assembly, narcissistic self-sorting, shape-dependent self-recognition, terpyridine

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